

IoT Water pollution Monitoring in Rivers and Lakes using RFsignal-controlled RC Boat

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Abstract— Ensuring water safety is a significant challenge due to the abundance of pollutants, many of which are caused by human activities. To Challenge the Industrial growth, agriculture, and lax regulations have polluted our water sources. Water quality problems arise primarily from the over-exploitation of natural resources. Uneven rainfall patterns and individual habits worsen the problem. Public confidence in water quality is declining due to concerns about pollution and aging infrastructure. This innovative project tackles the crucial first step to monitoring water pollution. It utilizes a cost-effective, time-saving remote-controlled (RC) boat equipped with sensors to collect readings from various water points. The boat is equipped with sensors that measure like pH level and turbidity, water quality. These readings are transmitted wireless to a remote control with an LCD screen for easy viewing. This technology empowers to effectively monitor water quality. By gathering data on contamination levels. This project introduces a practical and user-friendly solution for assessing water quality. By combining readily available RC boat technology with specialized sensors, it offers a valuable tool for environmental monitoring.

Keywords—monitor the quality of water, pH and Turbidity, a remote-controlled boat with sensors, LCD display.

I. INTRODUCTION

Improving Technological advancements are proving to be valuable allies in the battle against environmental challenges. Recently, To monitoring water pollution in reservoirs, rivers, and lakes. Concerns among the public regarding the safety of tap water have been growing. There is mounting anxiety about water contamination and its potential impact on drinking water sources.

The key to addressing water pollution lies in accurately quantifying its direction. Reliable data on water quality sets the stage for effective pollution mitigation, ultimately restoring public trust in the safety of tap water and contributing to the creation of a healthier environment. This patent introduces a system that tackles this issue by measuring indicators of water quality, pH and turbidity. Turbidity readings serve as a valuable tool for understanding pollution levels, information for decision-making for water treatment and pollution control strategies. This innovative technology empowers to managing water quality. By providing precise measurements, it serves as an early warning system for potential contamination incidents, prompting authorities to take immediate corrective actions.

II. Proposed Work

The main objective of the paper is to provide a cost effective , time efficient and dynamically reliable method for measuring the water quality using a RC boat which can measure the pH and Turbidity level and give us the contamination level of the water bodies. Project contains a remote controlled boat with sensors. Users can drive the boat using a remote and the boat will collect all readings after a specific interval. All the readings measured by sensors will be sent to the remote and can be seen on the LCD display attached to the remote. This Project will be really beneficial for human beings to monitor the Water quality and take adaptive measures for it.

Water Quality Monitoring is important to ensure that it is safe for humans ,wildlife and marine life to drink.It is important to measure water quality to understand environmental impacts and to not harm sea life. In our Project we have planned to process our invention in two process, First process is to design a Radio Controlled Boat with Servo motor and propeller shaft and the Second process is to interface pH sensor, Turbidity sensor, Temperature sensor and the LCD display with the Arduino and also interface the Rf receiver Transmitter to the Arduino. The potential of Hydrogen (pH) is a measurement that indicates the concentration of hydrogen ions in water, thereby determining its acidity or alkalinity. The pH scale ranges from 0 to 14, with 0 representing high acidity, 14 high alkalinity, and 7 being neutral. This measurement is critical for understanding water quality, especially in relation to the survival of amphibians and other aquatic organisms. Water with a pH value between 6.5 and 7.5 is considered safe for drinking, while a pH lower than 4 can be harmful to aquatic life.

Turbidity refers to the clarity of water, indicating the presence of particles like plants, clay, silt, and debris. Excessive turbidity can have detrimental effects on the reproductive success of aquatic species, as sediment can cover spawning areas and impact the viability of eggs. Turbidity is commonly measured in Nephelometric Turbidity Units (NTU), providing a standardized metric for assessing water clarity. Understanding and monitoring these parameters, including temperature, turbidity, and conductivity, are essential for comprehensive water quality assessment and management..

A. Designing of RC Boat:

The main purpose of our project is to monitor the quality of water in lakes, reservoirs, rivers, etc. We cannot monitor water quality in such big oceanic rivers by using a stationary

system because water parameters vary at every single location. To avoid this, we manufactured a boat that can float and move on the water simply with a user controller. Our structure is designed as a hull shape, which reduces the resistivity of water flow and maintains the stability of the water. We have designed a radio-controlled boat using metal sheets for floating purposes. The movement of the RC boat is controlled by the RF receiver and Rf transmitter. We have configured our RC boat in a way that, According to the Command given by the Rf Transmitter, the Command is Transmitted to the RF receiver attached to the RC boat, and then the Signal is forwarded to the servo motor attached to the Boat from the Direction and Flow movement of the boat takes place. The Propeller shaft and Rudder assist the Servo motor for better Floating movement of the Boat. Initially, we designed a 3D diagram of the RC Boat for fabrication. In our RC boat, we are using a Two-channel RF receiver-transmitter to send signals to two servo motors; we are using two 500 rpm 12V servo motors for both side directions and forward and reverse movement. We have designed a 3D model of the RC boat using CAD software, and we have attached the 3D model in Figure 1.

A. Interfacing of water pollution monitoring sensors:

For configuring the Water contamination (Pollution) we have chosen to measure two parameters such as pH and Turbidity to measure the Water pollution level. We are using a pH sensor and Turbidity sensor to monitor the pH level and Turbidity level of the water bodies. The Turbidity sensor gives the output in Nephelometric Turbidity Units. pH sensor gives the reading in logarithmic units, if the value is with 0-7 then it comes under Acidic nature whereas if it lies within 7-14 it comes under Basic nature. By measuring those parameters and combining it, we give the Water pollution level and it instantly displays the water pollution level in the LCD display attached in the RC boat. This is configured interfacing the all the Sensors, LCD with the Arduino and giving connection supply from a 12V battery to Arduino.

After completion of the two phases of work the Water pollution monitoring system is integrated with the RC boat and the Connection from the RC boat such as DC motor, RF receiver are interfaced to the Arduino and now by combining them we acquire the fully working model of IOT water Pollution Monitoring using RC Boat. Now when the project is placed in the water bodies the movement and Direction of RC boat is controlled by the RF Transmitter operated by the user and also at the same time the Water pollution monitoring system in the RC boat measures the pH, Turbidity level and Temperature for the first time and after 10 seconds it measures the values again and this process is repeated again.

The main objective of the invention is to provide a cost-effective, time-efficient, and dynamically reliable method for measuring water quality using an RC boat, which can measure the pH and turbidity level and determine the contamination level of the water bodies. The project contains a remote-controlled boat with sensors. Users can drive the boat using a remote, and the boat will collect all readings after a specific interval. All the readings measured by sensors will be sent to the remote and can be seen on the LCD display attached to the remote. This Project will be really beneficial for human beings

to monitor the Water quality and take adaptive measures for it.

Figure 1 briefly describes the planning and 3D design of our RC Boat. We planned the dimensions of our Boat and designed it in SolidWorks software. According to the dimensions in the Designs, we started Fabricating the Boat model using Mild steel.

Figure 2 we have attached the Fabricated model of the RC boat. According to the Design we have fabricated our RC Boat using Mild steel. Since the RC boat was too heavy for floating purposes. So we have attached sponge like components in the sidelines in order to float. In the rear we have attached a Propeller Shaft for the movement of the RC boat in water. The Propeller shaft is also made of mild steel material.



Fig.2 Fabricated model of RC boat

Figure 3 There are many components and sensors in the Water quality monitoring system so it is difficult to insert all the components inside the RC boat which will not give us a clear ergonomics. So we have attached a plywood on top of the RC Boat which consists of water quality monitoring sensors and components. The Water quality monitoring system in the plywood consists of pH sensor, Turbidity sensor, Temperature sensor, Micro controller 8 (Arduino), LCD display, Rf receiver, Transistor and a switch. We have interfaced all these and derived the water quality monitoring system on the Plywood which will be attached on the top of the RC boat.

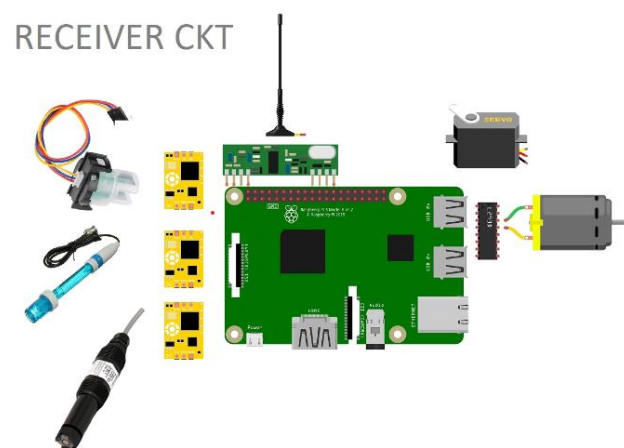


Fig.3 Water quality monitoring sensors interfacing

Figure 4 we have accomplished our project objective and have derived the water quality parameters such as pH and turbidity. The values are displayed in the LCD display. Now we have achieved the IOT water quality monitoring system using RC Boat. The Movement of the RC boat is controlled by the RF transmitter.



Fig.4 Working model of the project

III. Results and Discussions

The results and discussions section of a research report on IoT water pollution monitoring in rivers and lakes utilizing RF signal operated remote control boats would normally analyze their implications in the results. Data Collection on Water Quality: Describe the information gathered by the IoT sensors on the remote-controlled boat, taking into account variables like conductivity, temperature, turbidity, pH levels, and dissolved oxygen. Pollution's Spatial Distribution: Provide information on the geographical distribution of pollution in the target water bodies, taking into account any notable differences between various sites. Temporal Variations: Talk about any daily or seasonal fluctuations in the water quality measures that were seen throughout the course of the investigation. .

REFERENCES

- [1] Hall, J.; Zaffiro, A.D.; Marx, R.B.; Kefauver, P.C.; Krishnan, E.R.; Haught, R.C.; Herrmann, J.G. On-line water quality parameters as indicators of distribution system contamination. *J. Am. Water Works Assoc.* 2007, 99, 66–77
- [2] Chung, W.Y.; Chen, C.L.; Chen, J.B. Design and implementation of low power wireless sensor system for water quality monitoring. In *Proceedings of the 2011 5th International Conference on Bioinformatics and Biomedical Engineering*, Wuhan, China, 10–12 May 2011
- [3] Lambrou, T.P.; Anastasiou, C.C.; Panayiotou, C.G.; Polycarpou, M.M. A low-cost sensor network for real-time monitoring and contamination detection in drinking water distribution systems. *IEEE Sens. J.* 2014, 14, 2765–2772.